

Muhammad Zain Altaf

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PERSONAL STATEMENT

A self-driven undergraduate of Computer Science with a grounded curiosity in computational analysis. I am enthusiastic about keeping up to date with advancements within the industry. I have industry experience in an agile development team focussing primarily on delivery and quality assurance. I am dexterous in the role I play within any team; having taken leadership roles but also contributing as a team member, eagerly collaborating with other members.

EDUCATION

University of Leicester - BSc Computer Science (Year 2&3 - Complete) 2024-2026

Relevant Modules:

- Operating Systems and Networking, Databases and Domain Modelling, Software Architecture and System Development I & II, Internet and Cloud Computing, Agile Group Projects + others

University of Manchester - BSc Computer Science (Year 1) 2021-2022

Relevant Modules:

- Mathematical techniques for Computer Science, Data science, Team project + others.

The West Bridgford School - A-Levels and GCSEs 2019-2021

- A* in Maths, A in Computer Science and Chemistry.
- 11 GCSEs with grades 9-6 including grade 9 in Mathematics and 8 in Computer Science.

WORK EXPERIENCE

NUDG - Software Quality Assurance April 2023 - March 2024

- Developed a live iOS and Android application with 7000+ registered users, creating hundreds of bookings per week totalling 30000+ taxi bookings and counting.
- Operated within an **Agile** development team using the **Scrum** methodology to create a super-app gaining experience in industry-standard softwares such as **Teams, Outlook, Jira, Confluence, Git** and more.
- Conducted manual **testing** based on identified issues and constructed alternative flows as fixes.

UoL Computer Science Society - Events Coordinator November 2024 - June 2025

- Collaborated with industry leader IBM to provide in depth sessions on using IBM Cloud.
- Organised and carried out a hackathon with over 100 people.
- Arranged the CMS Ball - a formal social event in conjunction with actuarial science and mathematics societies.

Office Shoes - Sales Assistant 2019-2021

- Delivered excellent **customer service** through product knowledge, styling advice, and efficient transaction handling while supporting stock management and visual merchandising standards.

Age Concern Charity - Weekly Volunteer 2019-2021

- Supported daily shop operations as cashier and stock assistant, engaging warmly with a diverse range of customers and community members.

PROJECTS

Carbon-Aware Workload Shifting and Optimisation - Final Year Dissertation

- Designed and built a discrete-event simulation engine in **Python** to evaluate four carbon-aware workload scheduling policies when adjusting environmental factors (e.g. season, workload types).
- Ingested and cleaned HPC workload trace (306,000+ tasks) and 5-minute resolution carbon intensity data (105,120 samples) to feed into the simulation engine in a controlled manner - via Jupyter .
- Analysed simulation performance and created a recommendation matrix to showcase trends in policy efficacy within different seasons.
- Created simulated systems to mimic real world cloud architecture - including a modular policy system and reserved/on demand/spot instance capacity model.

Metropolitan Police Crime Prediction - Big Data Coursework

- Cleaned and reduced a 1.13 million-row police crime dataset through deduplication, outlier removal, and missing-value handling, then aggregated incidents into 58,353 LSOA-by-month records using a **MapReduce-style** pipeline.
- Engineered temporal features including lag-1 and rolling-mean predictors, and applied **K-Means clustering** with elbow and silhouette diagnostics to generate spatial context features.
- Built and evaluated **Linear Regression** and **Random Forest** models against a naive baseline using a chronological train-test split and **TimeSeriesSplit** cross-validation, achieving $R^2 = 0.94$ on held-out data.
- Produced an analytical report with exploratory visualisations, correlation analysis, and critical evaluation of model performance and limitations.

IBM Learning Platform - University Group Project

- Developed and integrated **RESTful APIs** using **Java Spring Boot**, ensuring seamless communication with a React/Next.js frontend.
- Contributed to a **MySQL-backed backend system**, implementing features such as gamification (streaks, leaderboards, badges) with attention to scalability and security.
- Acted as **Scrum Master**, coordinating daily stand-ups, managing the task board, and facilitating collaboration to keep development on track.

SKILLS AND INTERESTS

Skills

- Strong understanding of **Git** and workflows surrounding collective development.
- Confident using industry standard collaboration technologies e.g. **Teams, Outlook, Jira, Confluence** and more.
- Ability to **multitask**, actively listening to ideas and feedback to produce an effective product.
- Knowledge and understanding of **core programming principles** with awareness of implementation within different languages.
- Ability to diagnose problems and trace core causes.

Interests

- Competed at multiple hackathons including The Reply Code Challenge 2022, UKIEPC Programming Contest 2023, Morgan Stanley Coding Challenge 2024 and Lets Hack Leicester 2024.